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# Carbapenem Resistant *Serratia marcescens* Harboring *bla*$_{OXA-48}$, *armA* and *aac*(6′)-Ib-cr Genes: The First Baseline Study from Pakistan

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## Abstract
The emergence of *Serratia marcescens* as a significant healthcare-associated pathogen is compounded by its rising carbapenem resistance resulting in a critical global health challenge. This study is the first in Pakistan to comprehensively characterize the antimicrobial resistance patterns and molecular mechanisms of *S. marcescens* from clinical specimens. The *S. marcescens* isolates were obtained from clinical specimens collected at tertiary care hospitals in Lahore and Faisalabad, Pakistan, between October 2023 and September 2024. Isolates were identified using the Vitek-2 system and MALDI-TOF mass spectrometry. The disk diffusion assays, and broth microdilution method were used for antimicrobial susceptibility testing. The antimicrobial resistance determinants were amplified by PCR followed by sanger sequencing. The study revealed concerning resistance rates among *S. marcescens* isolates: 78% were resistant to third-generation cephalosporins, 52% to carbapenems, and 71.6% to fluoroquinolones. However, no isolates were found to be resistant to the tigecycline. The carbapenem resistance genes were distributed among carbapenem-resistant *S. marcescens* (CRSM) strains as follows: *bla*$_{OXA-48}$ was most prevalent at 69.2%, followed by *bla*$_{KPC}$ at 20.8%, and *bla*$_{IMP}$ at 10% of isolates. The 16S methylase gene *armA* was found exclusively in CRSM strains, present in 73.1% (95/130) of isolates. The *aac*(6′)-Ib-cr gene variant, which confers resistance to aminoglycosides and fluoroquinolones, was found in 37.2% of CRSM isolates. The CRSM isolates co-harboring *bla*$_{OXA-48}$, *armA*, and aac(6′)-Ib-cr genes demonstrated a convergence of carbapenem, aminoglycoside, and fluoroquinolone resistance mechanisms which confer a multi-drug resistance phenotype in these isolates, with tigecycline remaining the sole susceptible agent. These findings expose an escalating AMR crisis in Pakistan driven by high-risk resistance gene combinations among *S. marcescens* isolates. This underscores the dire necessity for urgent nationwide interventions including rapid diagnostics, genomic surveillance of resistance mechanisms, and strict antibiotic stewardship to curb the increasing antimicrobial resistance threats and nosocomial transmission of such pathogens.

## Introduction

*Serratia marcescens* is a Gram-negative rod-shaped bacterium that belongs to the *Enterobacteriaceae* family. *S. marcescens* has evolved from an occasional opportunistic pathogen to a significant cause of hospital infections, becoming a serious healthcare threat that demands considerable attention from the global medical community [1, 2]. The increasing frequency of *S. marcescens* infections in clinical settings led the World Health Organization (WHO) to classify it as a priority antibiotic-resistant pathogen in 2017 [3]. Surveillance data from the European Center for Disease Prevention and Control (ECDC) has positioned *Serratia* species as a significant healthcare threat, ranking sixth among bacteria causing hospital-acquired pneumonia and ninth for both bloodstream and urinary tract infections [4].

*S. marcescens* exhibits intrinsic resistance to several antimicrobial agents, including narrow-spectrum penicillins (e.g., ampicillin), first- and second-generation cephalosporins, cephamycins, macrolides, and polymyxins, and is therefore considered a significant concern in clinical settings. [5–7]. This intrinsic resistance to the penicillins and cephalosporins is primarily due to the production of chromosomally encoded AmpC β-lactamases while typically leaving fourth-generation cephalosporins and carbapenems as preferred therapeutic options [6, 7]. However, recent studies

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from various parts of world have demonstrated increasing resistance to carbapenems among *S. marcescens* isolates [4, 6, 8, 9]. Such findings raise significant concern as carbapenems are considered the most potent treatment option for Gram-negative bacteria especially if they are resistant to third and fourth-generation cephalosporins. The worldwide emergence of carbapenem-resistant Gram-negative bacteria represents a major challenge for global health systems. This resistance pattern is primarily attributed to the ability of bacteria to produce carbapenemases which are considered as the most powerful group of beta-lactamases [10–12]. These acquired beta-lactamases conferring carbapenem resistance among *S. marcescens* exhibit considerable variability across different studies. This variability is influenced by regional factors, the specific strains involved, and the mechanisms of resistance identified.

Despite its global recognition as a critical pathogen, no studies on the AMR among *S. marcescens* exist in Pakistan. To the best of our knowledge, this is the first comprehensive study investigating *S. marcescens* in Pakistan. A recent case study reported a pandrug-resistant (PDR) strain of *S. marcescens* isolated from a urinary culture [13]. However, there remains a critical gap in understanding the broader landscape of resistance and underlying mechanisms for *S. marcescens* across clinical settings in Pakistan. This study aims to address the knowledge gaps by conducting a comprehensive phenotypic and molecular characterization of *S. marcescens* isolated from multiple tertiary care hospitals in Lahore and Faisalabad, providing the first extensive investigation into the resistance patterns, antimicrobial susceptibility, and molecular mechanisms of resistance in *S. marcescens* from clinical specimens in the region. In this study, we have provided the first national prevalence data for carbapenem resistance and associated genes. Moreover, this work reports the first evidence of CRSM strains co-harboring $bla_{\text{OXA-48}}$, *armA*, and aac(6′)-Ib-cr. This study has established a baseline for monitoring AMR among *S. marcescens* in Pakistan and the foundational insights into regional AMR mechanisms to guide empirical therapy and stewardship policies.

## Materials and Methods

### Clinical Isolates

This retrospective study analyzed *S. marcescens* isolates recovered from clinical specimens of hospitalized patients at the tertiary care hospitals in Lahore and Faisalabad, Pakistan, between October 2023 and September 2024. Inclusion criteria comprised all non-duplicate *S. marcescens* isolates identified during routine diagnostic workflows, excluding environmental or surveillance samples. Exclusion criteria removed duplicate isolates from the same patient. Demographic and clinical data (age, gender, specimen type) were extracted retrospectively from the medical records.

### Isolation and Identification

All samples except for urine specimens were cultured on blood, chocolate, and MacConkey agar plates (Oxoid, Basingstoke, UK). The urine samples were cultured on cystine-lactose-electrolyte-deficient (CLED) medium (Oxoid, Basingstoke, UK). The inoculated agar plates were incubated aerobically at a temperature of 37 ± 1 °C for a duration of 24–48 h. The isolates were identified through examination of colonial morphology, Gram staining, and biochemical assays, followed by Vitek-2 system (bioMérieux, France). The *S. marcescens* isolates were further confirmed by bioMérieux VITEK® Mass Spectrometry (MS) Matrix-Assisted Laser Desorption Ionization-Time of Flight (MALDI-TOF) after appropriate pre-treatments as per manufacturer guidelines. For long-term preservation, the bacterial strains were maintained in brain heart infusion (BHI) broth (Oxoid, Basingstoke, UK) with 40% glycerol at −80 °C.

### Antimicrobial Susceptibility Testing

Antimicrobial susceptibility testing (AST) was performed using both disk diffusion and broth microdilution methods to ensure comprehensive resistance profiling. Initially, the disk diffusion assays were conducted for the AST followed by broth microdilution method to measure the minimum inhibitory concentration (MIC) values for critical antibiotics (e.g., carbapenems, tigecycline). In cases of discordant results between methods, both tests were repeated and if discrepancies persisted, the MIC data were selected for final interpretation.

Disk diffusion tests were conducted for the following antimicrobial agents: amikacin (AK, 30 µg), gentamicin (CN, 10 µg), tobramycin (TOB, 10 µg), imipenem (IPM, 10 µg), meropenem (MEM, 10 µg), ceftazidime (CAZ, 30 µg), cefotaxime (CTX, 30 µg), cefixime (CFM, 5 µg), cefepime (FEP, 30 µg), piperacillin-tazobactam (TZP, 100/10 µg), doxycycline (DO, 30 µg), ciprofloxacin (CIP, 5 µg), levofloxacin (LEV, 5 µg), and trimethoprim-sulfamethoxazole (SXT, 1.25/23.75 µg). Antimicrobial disks were sourced from Oxoid, UK.

The MICs for nine antibiotics, including CTX, CRO, IPM, MEM, AK, CN, CIP, SXT, and tigecycline (TGC), were determined using broth microdilution. MICs were interpreted according to CLSI 2024 guidelines [14], except for TGC, for which Food and Drug Administration (FDA, USA) criteria were applied: susceptible (MIC $\leq 2$ µg/mL), intermediate (MIC $= 4$ µg/mL), and resistant (MIC $\geq 8$ µg/mL) [10]. *Escherichia coli* (ATCC® 25,922 and ATCC®

35,218) and *Pseudomonas aeruginosa* ATCC® 27,853 were employed as quality control strains.

### Antimicrobial Resistance Determinants

The bacterial DNA of *S. marcescens* isolates was extracted using a bacterial DNA extraction kit (FavorPrep™, Favorgen Biotech Corp., Ping Tung, Taiwan) according to the manufacturer's instructions. DNA was stored at − 70 °C for further analysis. The purity of DNA was determined via absorbance measured using Nano-Drop (Thermo Fisher Scientific, Cambridge, UK) at wavelengths of 260 and 280 nm.

The *S. marcescens* isolates were screened for antimicrobial resistance genes using primers listed in Table 1. The analysis targeted genes associated with key resistance mechanisms, including extended-spectrum β-lactamases (ESBLs: $bla_{CTX-M}$, $bla_{TEM}$, and $bla_{SHV}$), class A carbapenemases ($bla_{KPC}$), metallo-β-lactamases (MBLs: $bla_{IMP}$, $bla_{VIM}$, $bla_{NDM}$, $bla_{SPM}$, $bla_{GIM}$, $bla_{SIM}$), and oxacillinases ($bla_{OXA-48}$). Additionally, aminoglycoside-modifying enzymes (AMEs: *aac*(3)-I, *aac*(3)-IIa, *aph*(3'')-Ib, and *ant*(3'')-Ia) and 16S rRNA methylases (*armA, rmtA, rmtB, rmtC, rmtD, rmtE,* and *rmtF*) were evaluated for aminoglycoside resistance. Plasmid-mediated quinolone resistance (PMQR) genes (*qnrA, qnrB,* and *qnrS*) and sulfonamide resistance genes (*sul1* and *sul2*) were also assessed. The *aac*(6')-Ib-cr gene variant was detected using PCR amplification targeting a 476 bp region. The reaction employed specific primers: Forward (5'-ATGACTGAGCATGACCTT G-3') and Reverse (5'-AACCATGTACACGGCTGG-3') as described previously [15].

PCR reactions were performed in a total volume of 25 µL using Green Taq Thermo Fisher Scientific, Cambridge, UK), master mix (2 ×) comprising Taq DNA polymerase, dNTPs, $MgCl_2$ (1.5 mM final concentration), reaction buffer, and tracking dye. Each reaction contained 12.5 µL of Green Taq master mix, 0.2 µM each of forward and reverse primers, and 10–50 ng of template DNA, with the remaining volume adjusted to 25 µL using nuclease-free water. Thermal cycling conditions included an initial denaturation at 95 °C for 3 min, followed by 35 cycles of denaturation (95 °C for 30 s), annealing (Variable temperature (Table 1) for 30–45 s) and extension (72 °C for 30 s to 1 min). A final extension step at 72 °C for 5 min was included, followed by indefinite holding at 4 °C. Negative controls having nuclease-free water instead of template DNA, were included in all experiments. PCR products were analyzed by agarose gel electrophoresis using (1.2% w/v) agarose gel in 1 × TAE buffer (pH 8.3) and ethidium bromide (0.5 µg/mL final concentration). The DNA bands were visualized under UV light. Following amplification, the PCR products were purified using a GeneJET PCR Purification Kit Thermo Fisher Scientific, Cambridge, UK), and sequenced by Macrogen (Seoul, South Korea). Sequence analysis was performed using the (BLAST, NCBI) tool.

## Results

### Demographic and Clinical Characteristics

A total of 250 non-duplicate *S. marcescens* isolates were collected from the clinical specimens obtained from patients admitted to tertiary care hospitals in Lahore and Faisalabad, Pakistan during October 2023 to September 2024. Of the isolates, 57.2% (143) were from male patients, and 42.8% (107) were from female patients. The age distribution showed that 42.8% (107) of the isolates were from patients aged 0–20 years, followed by 23.6% (59) from those aged 21–40 years, 20.8% (52) from those aged 41–60 years, and 12.8% (32) from patients older than 60 years.

Blood samples were the most common source of *S. marcescens* isolates, comprising 45.2% (113/250) of the total isolates, followed by pus specimens, which accounted for 24.8% (62) of the isolates, and urine samples, representing 14% (35) (Table 2).

The majority of *S. marcescens* isolates (74.4%) were recovered as the sole organism in cultures, though co-isolation with other microorganisms was also observed. *P. aeruginosa* was the most frequently co-isolated organism (6.4%), followed by coagulase-negative Staphylococci (5.6%) and *E. coli* (4.0%). *K. pneumoniae*, *Enterococcus* spp., and *S. aureus* were also identified alongside *S. marcescens* at lower frequencies as shown in Fig. 1.

### Antimicrobial Resistance of *S. marcescens* Clinical Isolates

Antimicrobial susceptibility testing of *S. marcescens* isolates revealed concerning resistance patterns across multiple antibiotic classes, posing significant therapeutic challenges. Resistance to third-generation cephalosporins (cefotaxime, ceftriaxone, and cefixime) was observed in 78% of isolates. 52% of the isolates were found resistant to carbapenems (imipenem and meropenem), and 71.6% to fluoroquinolones (ciprofloxacin and levofloxacin). Among aminoglycosides, resistance rates varied: amikacin had the lowest rate at 49.6%, whereas gentamicin and tobramycin exhibited resistance rates of 68.8% and 65.6%, respectively. The resistance to trimethoprim-sulfamethoxazole was 62.0% whereas all the tested isolates were found susceptible to tigecycline (100%) (Fig. 2).

**Table 1** Primers used in this study

| Genes | Primers | Sequence (5' to 3') | Aimed product (bp) | Annealing temperature (°C) | References |
|---|---|---|---|---|---|
| *bla*$_{TEM}$ | TEM-F | TCAACATTTCCGTGTCG | 860 | 56 | [44] |
| | TEM-R | CTGACAGTTACCAATGCTTA | | | |
| *bla*$_{SHV}$ | SHV-F | ATGCGTTATATTCGCCTGTG | 896 | 56 | |
| | SHV-R | AGATAAATCACCACAATGCGC | | | |
| *bla*$_{CTX-M}$ | CTXMU-F | ATGTGCAGYACCAGTAARGTKATGGC | 593 | 52 | [45] |
| | CTXMU-R | TGGGTRAARTARGTSACCAGAAYCAGCGG | | | |
| *bla*$_{OXA-48}$ | OXA-48-F | TTGGTGGCATCGATTATCGG | 744 | 52 | [46] |
| | OXA-48-F | GAGCACTTCTTTTGTGATGGC | | | |
| *bla*$_{IMP}$ | IMP-F | GGAATAGAGTGGCTTAAYTC | 232 | 50 | [45] |
| | IMP-R | TCGGTTTAAYAAAACAACCACC | | | |
| *bla*$_{VIM}$ | VIM-F | GATGGTGTTTGGTCGCATA | 390 | 50 | |
| | VIM-R | CGAATGCGCAGCACCAG | | | |
| *bla*$_{SPM}$ | SPM-F | AAAATCTGGGTACGCAAACG | 271 | 52 | [47] |
| | SPM-R | ACATTATCCGCTGGAACAGG | | | |
| *bla*$_{GIM}$ | GIM-F | TCGACACACCTTGGTCTGAA | 477 | 50 | |
| | GIM-R | AACTTCCAACTTTGCCATGC | | | |
| *bla*$_{SIM}$ | SIM-F | TACAAGGGATTCGGCATCG | 570 | 50 | |
| | SIM-R | TAATGGCCTGTTCCCATGTG | | | |
| *bla*$_{NDM}$ | NDM-F | CACCTCATGTTTGAATTCGCC | 984 | 52 | [46] |
| | NDM-R | CTCTGTCACATCGAAATCGC | | | |
| bla$_{KPC}$ | KPC-F | CGTCTAGTTCTGCTGTCTTG | 798 | 50 | |
| | KPC-R | CTTGTCATCCTTGTTAGGCG | | | |
| *arm*A | armA-F | ATTCTGCCTATCCTAATTGG | 315 | 56 | |
| | armA-R | ACCTATACTTTATCGTCGTC | | | |
| *rmt*A | rmtA-F | CTAGCGTCCATCCTTTCCTC | 635 | 56 | |
| | rmtA-R | TTTGCTTCCATGCCCTTGCC | | | |
| *rmt*B | rmtB-F | ATGAACATCAACGATGCCCT | 769 | 56 | |
| | rmtB-R | CCTTCTGATTGGCTTATCCA | | | |
| *rmt*C | rmtC-F | CGAAGAAGTAACAGCCAAAG | 711 | 56 | |
| | rmtC-F | ATCCCAACATCTCTCCCACT | | | |
| *rmt*D | rmtD-F | CGGCACGCGATTGGGAAGC | 401 | 51 | |
| | rmtD-R | CGGAAACGATGCGACGAT | | | |
| *rmt*E | rmtE-F | ATGAATATTGATGAAATGGTTGC | 818 | 46 | |
| | rmtE-R | TGATTGATTTCCTCCGTTTTTG | | | |
| *rmt*F | rmtF-F | GCGATACAGAAAACCGAAGG | 589 | 50 | |
| | rmtF-R | ACCAGTCGGCATAGTGCTTT | | | |
| *aac(3)-IIa* | *aac(3)-IIa-F* | GGCAATAACGGAGGCGCTTCAAAA | 563 | 58 | [48] |
| | *aac(3)-IIa-R* | TTCCAGGCATCGGCATCTCATACG | | | |
| *aac(3)-I* | *aac(3)-Ia-F* | GCAGTCGCCCTAAAACAAA | 464 | 52 | [49] |
| | *aac(3)-Ia-R* | CACTTCTTCCCGTATGCCCAACTT | | | |
| *aph*(3″)-Ib | aph(3″)-Ib-F | GTGGCTTGCCCCGAGGTCATCA | 612 | 55 | [48] |
| | aph(3″)-Ib-F | CCAAGTCAGAGGGTCCAATC | | | |
| *ant*(3″)-Ia | ant(3″)-Ia-F | TCGACTCAACTATCAGAGG | 245 | 50 | [50] |
| | ant(3″)-Ia-R | ACAATCGTGACTTCTACAGCG | | | |
| *qnrA* | *qnrA*-F | GGATGCCAGTTTCGAGGA | 492 | 52 | [51] |
| | *qnrA*-R | TGCCAGGCACAGATCTTG | | | |

**Table 1** (continued)

| Genes | Primers | Sequence (5' to 3') | Aimed product (bp) | Annealing temperature (°C) | References |
|---|---|---|---|---|---|
| *qnrB* | *qnrB*-F | GGMATHGAAATTCGCCACTG | 264 | 55 | [52] |
| | *qnrB*-R | TTTGCYGYYCGCCAGTCGAA | | | |
| *qnrS* | *qnrS*-F | GCAAGTTCATTGAACAGGGT | 428 | 50 | |
| | *qnrS*-R | TCTAAACCGTCGAGTTCGGCG | | | |
| *sul*1 | sul1-F | CGGCGTGGGCTACCTGAACG | 433 | 58 | [44] |
| | sul1-R | GCCGATCGCGTGAAGTTCCG | | | |
| *sul*2 | sul2-F | GCGCTCAAGGCAGATGGCATT | 293 | 58 | |
| | sul2-R | GCGTTTGATACCGGCACCCGT | | | |
| *sul3* | sul3-F | GAGCAAGATTTTTGGAATCG | 880 | 50 | [53] |
| | sul3-F | CATCTGCAGCTAACCTAGGGCTTTGGA | | | |

**Table 2** Specimen wise frequency distribution of *S. marcescens* isolates

| Specimen | Frequency *n* (%) |
|---|---|
| Blood | 113(45.2) |
| Pus | 62(24.8) |
| Urine | 35(14) |
| Tissue | 10(4) |
| CVP tip | 7(2.8) |
| Pleural fluid | 7(2.8) |
| CSF | 5(2) |
| Tracheal secretion | 2(0.8) |
| Vitreous aspirate | 2(0.8) |
| Ascitic fluid | 1(0.4) |
| Bronchial washing | 1(0.4) |
| Conjunctival swab | 1(0.4) |
| Ear swab | 1(0.4) |
| Peritoneal fluid | 1(0.4) |
| Sinus secretion | 1(0.4) |
| Synovial fluid | 1(0.4) |

## AMR Determinants in CSSM and CRSM Isolates

Among 120 CSSM isolates, resistance to cephalosporins was widespread, with 55.8% (67/120) resistant to cefixime, 54.2% (65/120) to cefotaxime and ceftriaxone, and 51.7% (62/120) to cefepime. Among these CSSM isolates, the ESBL genes $bla_{\text{CTX-M}}$, $bla_{\text{TEM}}$, and $bla_{\text{SHV}}$ were detected in 11.7%, 9.2%, and 2.5% isolates, respectively. For aminoglycosides, 39.2% (47/120) of isolates exhibited resistance to gentamicin, 32.5% (39/120) to tobramycin, and 12.5% (15/120) to amikacin. The *aac*(3)-I gene present in all gentamicin-resistant isolates (47/47) and *aac*(3)-IIa in 31.9% (15/47) of these isolates. Fluoroquinolone resistance was observed in 40.8% (49/120) of isolates for both ciprofloxacin and levofloxacin, with the *aac*(6')-Ib-cr resistance gene detected in 85.7% (42/49) of fluoroquinolone-resistant strains and *qnrB* in 18.4% (9/49) and *qnrS* in 6% (3/49) isolates. Additionally, 34.2% (41/120) of isolates were resistant to trimethoprim-sulfamethoxazole, where the *sul1* gene was universal (100%, 41/41) and *sul2* occurred in 14.6% (6/41) of resistant isolates.

Among 130 CRSM isolates, pan-resistance was observed to cephalosporins (cefixime, cefotaxime, ceftriaxone, cefepime), fluoroquinolones (ciprofloxacin, levofloxacin), and piperacillin-tazobactam (TZP). Widespread aminoglycoside resistance was noted, with 96.2% (125/130) of isolates resistant to gentamicin and tobramycin, and 83.8% (109/130) to amikacin. High resistance rates were also seen for trimethoprim-sulfamethoxazole (87.7%, 114/130) and doxycycline (77.7%, 101/130), though all isolates remained susceptible to tigecycline. Genotypic analysis identified the oxacillinase gene $bla_{\text{OXA-48}}$ as dominant, detected in 69.2% (90/130) of isolates, followed by class A beta-lactamase gene $bla_{\text{KPC}}$ (20.8%, 27/130) and the metallo-beta-lactamase gene $bla_{\text{IMP}}$ (10.0%, 13/130). The 16S rRNA methyltransferase gene *armA* was present in 73.1% (95/130) of isolates, and the quinolone resistance gene *aac*(6')-Ib-cr was identified in 39.2% (51/130) of isolates (Table 3).

## AMR Profile and Associated Genetic Determinants in S. marcescens Isolates

A comprehensive analysis of 250 *S. marcescens* isolates revealed distinct resistance profiles and associated resistance genes. The most prevalent multidrug-resistant pattern was observed in 90 isolates (36%), showing resistance to all tested antibiotics except tigecycline. These isolates harbored multiple β-lactamase genes ($bla_{\text{SHV}}$ (14/90), $bla_{\text{TEM}}$ (15/90) and $bla_{\text{CTX-M}}$ (13/90), with $bla_{\text{OXA-48}}$ present in all 90 isolates. These are also found to harbor the 16S methylase gene i.e. *armA* in 76 isolates. The second most common resistance pattern was observed in 13 isolates (5.3%) which

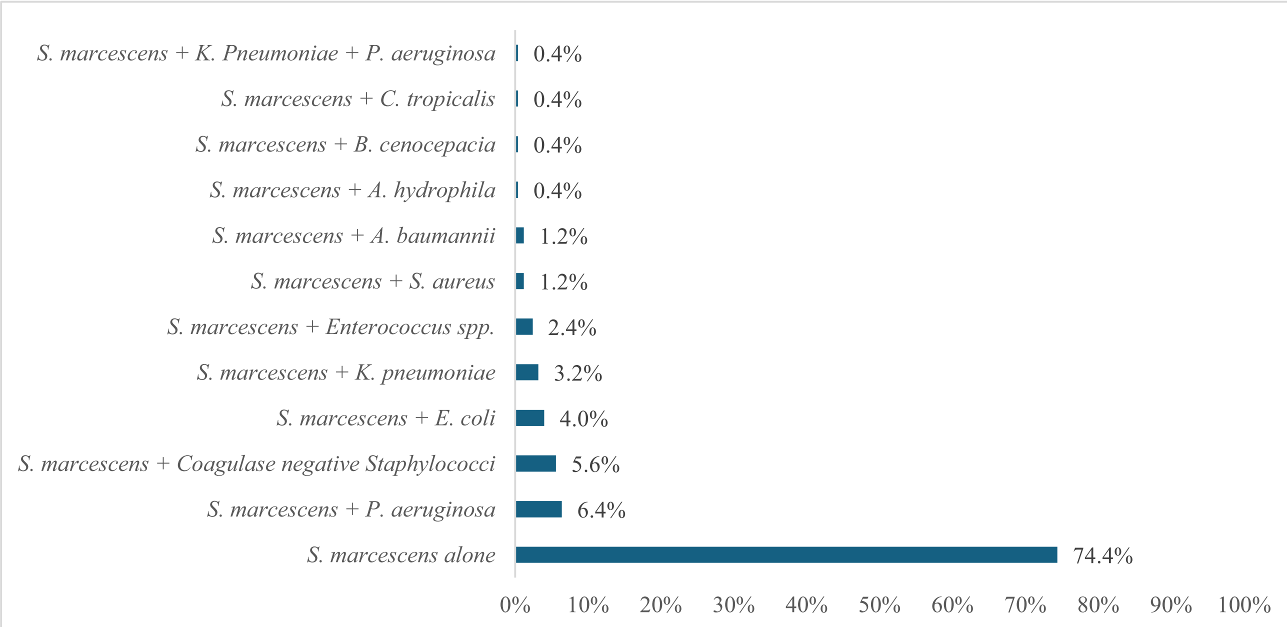


**Fig. 1** Distribution of *S. marcescens* isolates in various clinical specimens as a single pathogen or in co-infections

were resistant to all tested antibiotics except doxycycline. All these 13 isolates carried $bla_{\text{CTX-M}}$ and $bla_{\text{IMP}}$ genes, along with various aminoglycoside and fluoroquinolone resistance determinants in some of the isolates. Notably, 25 isolates (10%) were susceptible to all tested antimicrobial agents (Table 4).

## Discussion

The present study revealed several significant findings related to AMR in *Serratia marcescens* isolates from the tertiary care hospitals of Punjab, Pakistan. The AMR patterns of *S. marcescens* in Pakistani healthcare settings revealed a critical public health concern. Our study documented alarmingly high resistance rates across multiple antibiotic classes, with particularly concerning resistance to third-generation cephalosporins i.e. 78%. This resistance rate significantly exceeds those reported in recent international studies. A Bulgarian study of 180 *S. marcescens* isolates found a resistance rate of 49.4% to third-generation cephalosporins, with the highest resistance observed among isolates from urine samples [16]. Similarly, a Turkish study on pediatric bloodstream infections (BSIs) demonstrated high rates of ESBL production (57.8%) and carbapenemase production (40%) conferring resistance to third-generation cephalosporins and carbapenems, respectively [17]. The study from Brazil have reported 54 CRSM isolates collected from adult and neonatal intensive care units that were non-susceptible to beta-lactam antibiotics including third and fourth-generation cephalosporins with 92.6% resistance to tigecycline [5].

The alarming prevalence of carbapenem resistance observed in this study (52%, 130/250) represents a critical escalation of AMR in the healthcare settings of Pakistan. Notably, 36% (90/250) of isolates exhibited pan-resistance to all tested antibiotics except tigecycline. Globally, most studies on CRSM have involved limited isolate collections and focused primarily on elucidating resistance mechanisms, while research assessing the prevalence of carbapenem resistance in this pathogen remains scarce. This finding is completely in contrast with the global reports where CRSM remains relatively uncommon. The prevalence of carbapenem resistance among *S. marcescens* varies widely across regions, with some studies reporting rates as low as no resistance in certain bacterial populations [18]. In a study, a total of 396 *S. marcescens* isolates from various clinical samples at a University Hospital in Italy were analyzed during January 2015 to December 2022. The resistance to carbapenem rises from 4.2% for meropenem in 2025 to 7.5% in 2020 that shows a rising trend that might be attributed to increased antibiotic pressure and hospital-acquired infections [19]. The high prevalence of carbapenem resistance in our study is especially alarming given that carbapenems are often considered last-resort antibiotics for treating serious gram-negative infections. These findings necessitate an immediate revision of empirical antibiotic protocols in Pakistani hospitals: carbapenems and third-generation cephalosporins should no longer be used as first-line agents

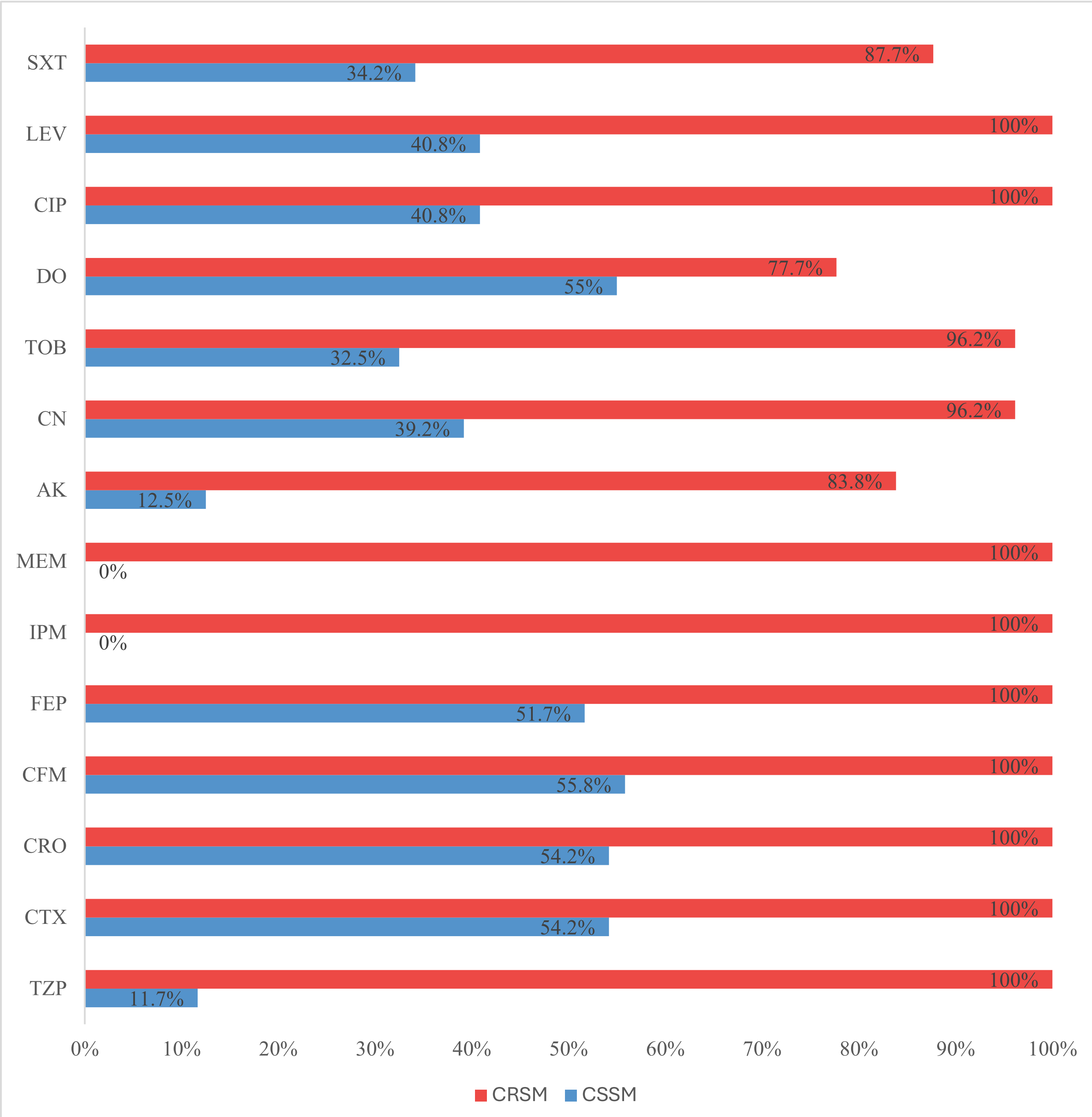


**Fig. 2** Overall antimicrobial susceptibility patterns of *S. marcescens* isolates

for suspected *S. marcescens* infections without confirmatory susceptibility testing.

This study provides the first large-scale baseline data from Pakistan revealing an alarming carbapenem resistance rate which underscores CRSM as a pervasive threat in this region. This prevalence focused approach can be helpful to fill the critical knowledge gaps by understanding the scale of resistance which is foundational for designing targeted infection control policies and stewardship interventions. This study highlights the position of Pakistan as a country within a high-AMR burden region likely to be intensified by the unregulated antibiotic use and limited stewardship. The near-universal resistance of CRSM isolates to β-lactams, fluoroquinolones, and aminoglycosides mirrors trends seen in *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *Acinetobacter baumannii* outbreaks especially in South Asia where clonal dissemination of carbapenemase resistant strains lead towards therapeutic failures [20].

This study highlights distinct differences in AMR profiles between CRSM and carbapenem-susceptible strains as the CSSM isolates retained susceptibility to third-generation cephalosporins (62%, 74/120), fluoroquinolones

**Table 3** Comparison of phenotypic resistance and genotypic determinants in carbapenem-susceptible and carbapenem-resistant *S. marcescens* isolates

| Antimicrobial class | Antimicrobial agents | Resistance phenotypes | | Resistance determinants | |
|---|---|---|---|---|---|
| | | CSSM *n* (%) | CRSM *n* (%) | CSSM (*n*, %) | CRSM (*n*, %) |
| Cephalosporins | Cefotaxime | 65 (54.2%) | 130 (100) | $bla_{\text{CTX-M}}$ (14, 21.5%) $bla_{TEM}$ (11, 16.9%) | $bla_{\text{CTX-M}}$ (31, 23.8%) $bla_{\text{TEM}}$ (21, 16.2%) *blaSHV* (30, 23.1%) |
| | Ceftriaxone | 65 (54.2%) | 130 (100) | | |
| | Cefixime | 67 (55.8%) | 130 (100) | | |
| | Cefepime | 62 (51.7%) | 130 (100) | | |
| Carbapenems | Imipenem | – | 130 (100) | – | $bla_{\text{OXA-48}}$ (90, 69.2%) $bla_{\text{KPC}}$ (27, 20.8%) $bla_{\text{IMP}}$ (13, 10.0%) |
| | Meropenem | – | 130 (100) | | |
| Aminoglycosides | Amikacin | 15 (12.5%) | 109 (83.8%) | *aac(3)-I* (47, 100%) *aac(3)-IIa* (15, 31.9%) *ant(3")-Ia* (4, 10.3%) | *aac(3)-I* (76, 58.5%) *aac(3)-IIa* (32, 24.6%) *ant(3")-Ia* (53, 40.8%) *armA* (95, 73.1%) |
| | Gentamicin | 47 (39.2%) | 125 (96.2%) | | |
| | Tobramycin | 39 (32.5%) | 125 (96.2%) | | |
| Fluoroquinolones | Ciprofloxacin | 49 (40.8%) | 130 (100) | *aac(6')-Ib-cr* (42, 85.7%) *qnrB* (9, 18.37%) *qnrS* (3, 6.1%) | *aac(6')-Ib-cr* (51, 39.2%) *qnrB* (22, 16.9%) *qnrS* (7, 5.4%) |
| | Levofloxacin | 49 (40.8%) | 130 (100) | | |
| Folate synthesis inhibitors | Trimethoprim-sulfamethoxazole | 41 (34.2%) | 114 (87.7%) | *sul1*(41, 100%) *sul2* (6, 14.6%) | *sul1* (114, 87.7%) *sul2* (13, 10.0%) |
| Tetracycline | Doxycycline | 66 (55%) | 101 (77.7%) | – | – |
| β-lactam/β-lactamase inhibitor | Piperacillin-tazobactam | 14 (11.7%) | 130 (100) | – | – |
| Glycylcyclines | Tigecycline | – | – | – | – |

(58.3%, 70/120), and aminoglycosides (51.7–72.5% susceptibility). These findings emphasize the critical need to differentiate CRSM from CSSM especially in developing countries like Pakistan as indiscriminate use of tigecycline risks may lead to resistance against this last-resort antibiotic [21]. To mitigate the development of resistance and to preserve the therapeutic options, antimicrobial stewardship programs must adopt tailored strategies: reserving tigecycline for CRSM infections lacking alternative treatments, while prioritizing cephalosporins, fluoroquinolones, or aminoglycosides for CSSM cases.

Most significantly, blood samples constituted the major source of *S. marcescens* isolates i.e. 45.2% (113/250). This high prevalence highlights the significant burden of invasive *S. marcescens* infections, which are associated with severe outcomes such as sepsis and mortality. Our findings align with a pediatric cohort study in Turkey, where all reported *S. marcescens* infections ($n = 45$) were BSIs over a 20-month period. The study identified central venous catheters (CVCs) as a critical risk factor for BSIs, with 43/45 patients having indwelling CVCs [17]. The substantial proportion of isolates from pus specimens (24.8%) and urine samples (14%) in our study further highlights the capacity of *S. marcescens* in causing diverse clinical infections such as wound and soft tissue infections as well as urinary tract infections. However, the predominance of BSIs underscores the need for stringent infection control in high-risk units (e.g., NICUs, ICUs) where invasive devices and prolonged hospitalization could amplify the risk of transmission.

The age distribution observed in our study highlights a distinct pattern as majority of isolates were recovered from 0 to 20-year-old age group. This predominance underscores *S. marcescens* as an emerging concern in pediatric and neonatal healthcare settings in Pakistan. This depicts the role of our hospitals as tertiary care referral centers for high-risk pregnancies and critically ill children. Moreover, most cases occurred in neonates in our study who are at a higher risk mainly due to prematurity, the use of invasive devices (e.g., ventilators, central catheters), or prolonged antibiotic exposure especially in neonatal ICUs. Such findings align with global reports of *S. marcescens* as a cause of BSIs and outbreaks in NICUs, often linked to contaminated medical equipment or antiseptics [22–24]. The declining prevalence of *S. marcescens* with advancing age contradicts previous studies, which indicate that healthcare-associated infections are more common in older patients due to comorbidities and prolonged hospitalization [5]. Hospitals must prioritize stringent infection control measures in pediatric units and antimicrobial cycling to reduce selection pressure. Furthermore, routine surveillance for CRSM colonization in patients with indwelling devices, along with contact isolation protocols, is essential to curb nosocomial transmission.

Another clinically significant observation was the frequent co-isolation of *S. marcescens* with other bacterial species. While most isolates (74.4%) were detected as

**Table 4** Distribution of resistance profiles and corresponding genetic determinants in 250 *Serratia marcescens* clinical isolates

| Category | Resistance profile | No. of isolates | Genotypes (*n*) |
|---|---|---|---|
| CRSM | TZP, CTX, CRO, CFM, FEP, IPM, MEM, AK, CN, TOB, DO, CIP, LEV, SXT | 90 | $bla_{SHV}$ (14), $bla_{TEM}$ (15), $bla_{CTX-M}$ (13), $bla_{OXA-48}$ (90), *aac*(3)-IIa (26), *aac*(3)-I (52), *aph*(3″)-Ib (11), *ant*(3″)-Ia (37), *armA* (76), *qnrB* (9), *qnrS* (7), *aac*(6′)-Ib-cr (19), *sul1* (90), *sul2* (13) |
| | TZP, CTX, CRO, CFM, FEP, IPM, MEM, AK, CN, TOB, CIP, LEV, SXT | 13 | $bla_{CTX-M}$ (13), $bla_{IMP}$ (13), *aac*(3)-I (8), *aph*(3″)-Ib (3), *ant*(3″)-Ia (5), *qnrB* (5), *aac*(6′)-Ib-cr (13), *sul1* (13) |
| | TZP, CTX, CRO, CFM, FEP, IPM, MEM, CN, TOB, DO, CIP, LEV, SXT | 11 | $bla_{SHV}$ (11), $bla_{KPC}$ (11), *aac*(3)-I (11), *ant*(3″)-Ia (11), *qnrB* (8), *aac*(6′)-Ib-cr (3), *sul1* (11) |
| | TZP, CTX, CRO, CFM, FEP, IPM, MEM, AK, CN, TOB, CIP, LEV | 6 | $bla_{TEM}$ (6), $bla_{KPC}$ (6), *aac*(3)-IIa (6), *armA* (6), *aac*(6′)-Ib-cr (6) |
| | TZP, CTX, CRO, CFM, FEP, IPM, MEM, CN, TOB, CIP, LEV | 5 | $bla_{CTX-M}$ (5), $bla_{KPC}$ (5), *aac*(3)-I (5), *aac*(6′)-Ib-cr (5) |
| | TZP, CTX, CRO, CFM, FEP, IPM, MEM, CIP, LEV | 5 | $bla_{SHV}$ (5), $bla_{KPC}$ (5), *aac*(6′)-Ib-cr (5) |
| CSSM | TZP, CTX, CRO, CFM, FEP, CN, TOB, DO, CIP, LEV, SXT | 10 | $bla_{TEM}$ (4), $bla_{CTX-M}$ (10), *aac*(3)-I (10), *aac*(6′)-Ib-cr (10), *sul1* (10) |
| | CTX, CRO, CFM, FEP, AK, CN, TOB, DO, CIP, LEV, SXT | 9 | *aac*(3)-IIa (9), *aac*(3)-I (9), *aac*(6′)-Ib-cr (9), *sul1* (9) |
| | TZP, CTX, CRO, CFM, FEP, CN, DO, CIP, LEV, SXT | 4 | $bla_{CTX-M}$ (4), *aac*(3)-I (4), *qnrB* (4), *sul1* (4) |
| | CTX, CRO, CFM, FEP, AK, CN, TOB, CIP, LEV, SXT | 3 | $bla_{TEM}$ (3), *aac*(3)-IIa (3), *aac*(3)-I (3), *aac*(6′)-Ib-cr (3), *sul1* (3) |
| | CTX, CRO, CFM, FEP, AK, CN, TOB, CIP, LEV | 3 | *aac*(3)-IIa (3), *aac*(3)-I (3), *aph*(3″)-Ib (3), *qnrS* (3) |
| | CTX, CRO, CFM, FEP, DO, CIP, LEV, SXT | 10 | *aac*(6′)-Ib-cr (10), *sul1* (10), *sul2* (6) |
| | CTX, CRO, CFM, FEP, CN, TOB, CIP, LEV | 5 | *aac*(3)-I (5), *qnrB* (5), *aac*(6′)-Ib-cr (5) |
| | CTX, CRO, CFM, FEP, CN, TOB, DO | 5 | *aac*(3)-I (5), *aph*(3″)-Ib (5) |
| | CTX, CRO, CFM, FEP, TOB | 4 | $bla_{TEM}$ (4), *ant*(3″)-Ia (4) |
| | CTX, CRO, CFM, FEP, CN | 3 | $bla_{SHV}$ (3), *aac*(3)-I (3) |
| | CTX, CRO, CFM, FEP | 11 | – |
| | CN, DO, CIP, LEV | 5 | *aac*(3)-I (5), *aac*(6′)-Ib-cr (5) |
| | DO, SXT | 5 | *sul1* |
| | DO | 18 | – |
| | Susceptible to all tested antimicrobial agents | 25 | – |

monocultures, the observed co-isolation particularly with *Pseudomonas aeruginosa* (6.4%) and coagulase-negative *Staphylococci* (5.6%) warrant careful interpretation. These findings may reflect as co-colonization especially in chronic wounds or respiratory niches where polymicrobial colonization is common [25]. Nevertheless, the presence of *P. aeruginosa* alongside *S. marcescens* might be clinically relevant, as both species exhibit intrinsic multidrug resistance and harbor mobile genetic elements that facilitate the exchange of resistance determinants [26, 27]. Such interactions could complicate therapeutic strategies by promoting synergistic resistance or cross-protection within biofilms, even in the absence of overt co-infection. This emphasizes the need for enhanced environmental disinfection in chronic wound and respiratory care settings, where polymicrobial biofilms may facilitate horizontal gene transfer of resistance determinants.

The molecular analysis of resistance determinants in our study provides crucial insights into the genetic basis of AMR in *S. marcescens* isolates. Notably, the oxacillinase gene $bla_{OXA-48}$ emerged as a critical contributor to carbapenem resistance, detected in 69.2% of CRSM strains. This finding aligns with global reports highlighting the rising prevalence of $bla_{OXA-48}$ producing CRSM, particularly in Mediterranean countries and the Middle East [28–30]. A study from South Africa comprised 21 CRSM isolates have reported $bla_{OXA-48}$ as the most common carbapenemase that was identified in 86% (18/21) of CRSM isolates [7]. The predominance of $bla_{OXA-48}$ is consistent with global trends where this oxacillinase dominate among carbapenem-resistant *Enterobacterales* [28]. Although endemic to regions such as South Asia, North Africa, and Turkey, $bla_{OXA-48}$ harboring pathogens are now being reported across Europe and South Africa and are linked to carbapenem treatment failures in infected patients [7, 31].

Among 130 CRSM isolates, 90 (69.2%) exhibited identical susceptibility profiles. The presence of $bla_{OXA-48}$ in all 90 isolates highlights the potential for clonal dissemination or horizontal gene transfer of carbapenemase-encoding plasmids as evident by the reported outbreaks caused by carbapenemase-producing *Enterobacterales*. For instance, an outbreak was reported in the United Kingdom during 2017–2018 which was caused by *bla*$_{OXA-48}$ producing *Enterobacterales* (predominantly *Klebsiella pneumoniae* and *Enterobacter cloacae*) [32]. While our study lacks whole-genome sequencing data to confirm genetic relatedness, the phenotypic uniformity of these isolates and the presence of *bla*$_{OXA-48}$ in 90 isolates underscore the urgent need for enhanced infection control measures and environmental disinfection to mitigate potential nosocomial transmission.

The distribution pattern of aminoglycoside-modifying enzymes (AMEs) revealed a complex landscape of resistance mechanisms. While AMEs are indeed widespread in Gram-negative bacteria, the studies have indicated that certain AMEs are intrinsic to this genus. For example genes encoding AMEs such as *aph(3″)* (conferring streptomycin resistance) and *aac*(6′)-Ic (associated with amikacin and tobramycin resistance) were detected in the majority of studied chromosomes whereas the *aad*A16 nucleotidyltransferase gene was identified in only four environmental strain genomes [33]. In our study, the *aac*(3)-I (49.2%) was predominated among the AMEs and correlated with the observed phenotypic resistance to aminoglycosides. The *aac*(3)-I enzymes are commonly found in *Enterobacterales* and non-fastidious gram-negative non-fermenters. The *aac*(3)-I enzymes confer resistance to gentamicin, sisomicin, and fortimicin [34]. The *aac*(3)-I gene in *Serratia* is an acquired resistance determinant as the genomic studies have highlighted the presence of *aac*(3)-I gene in multidrug-resistant (MDR) *Serratia* strains, often along with other mobile genetic elements [7].

The *aac*(6′)-Ib-cr gene variant was identified in 52% of fluoroquinolone-resistant *S. marcescens* isolates. This gene encodes a modified version of the AAC(6′)-Ib enzyme, which typically acetylates aminoglycosides. Unlike the original enzyme, the *aac*(6′)-Ib-cr variant also acetylates fluoroquinolones such as ciprofloxacin and norfloxacin, conferring low-level resistance to these antibiotics [35]. This dual functionality makes the AAC(6′)-Ib-cr enzyme a significant concern in managing infections caused by *S. marcescens* and other gram-negative pathogens. Notably, the *aac(6′)-Ib-cr* gene has been detected in diverse bacterial species, including *Klebsiella pneumoniae*, *Escherichia coli*, and *Pseudomonas* spp. [36]. Furthermore, this gene frequently co-occurs with other resistance determinants, such as ESBL encoded genes which confer β-lactam resistance and *qnr* genes (mediating fluoroquinolone resistance) [37]. Such genetic co-occurrence promotes a multidrug-resistant phenotype which greatly limits therapeutic options.

16S rRNA methylases are enzymes that confer high-level resistance to aminoglycosides in Gram-negative bacteria. By modifying the 16S rRNA, these enzymes significantly reduce the efficacy of aminoglycosides, complicating treatment options for infections caused by multidrug-resistant pathogens. Notably, the 16S rRNA methylase gene *armA* was identified in 38% of *Serratia marcescens* isolates in this study and was exclusively associated with CRSM isolates (73.1%), explaining the observed aminoglycoside resistance phenotypes. Previous studies highlight the global prevalence and clinical impact of *armA*. For example, a cluster of 15 *S. marcescens* isolates co-harboring extended-spectrum β-lactamase (ESBL) genes and the *armA* methyltransferase was reported in a urology unit in Eastern Algeria (2011–2013), among the 54 clinical isolates analyzed in the study [38]. Another study demonstrated that 86% (18/21) of CRSM isolates were co-harboring *armA* and *bla*$_{OXA-48}$, underscoring its role in pan-resistant CRSM strains [7]. Other 16S rRNA methylases such as *rmtB*, *rmtA*, *rmtC*, *rmtE*, *rmtD*, and *rmtF* were not detected in our study. This contrasts with findings from China, where *rmtB* was reported alongside *aac*(6')-Ib-cr in resistant strains [39].

The *sul1* gene emerged as the most prevalent resistance determinant overall, present in 62% of isolates, with a significantly higher prevalence in CRSM (87.7%) compared to CSSM isolates (34.2%). This high prevalence correlates strongly with the observed phenotypic resistance to trimethoprim-sulfamethoxazole (62.0%). The *sul1* gene facilitates antibiotic resistance through antibiotic target replacement, allowing bacteria to survive in the presence of sulfonamide drugs. The detection of *sul1* raises concerns about co-selection of resistance genes, as this determinant is frequently embedded within class 1 integrons which usually maintain multiple resistance cassettes, even in the absence of direct selective pressure for sulfonamides [40, 41].

The high prevalence of multidrug resistance observed in our study poses significant therapeutic challenges in managing *S. marcescens* infections. Notably, 36% of isolates demonstrated resistance to all tested antibiotics except tigecycline, drastically limiting treatment options and underscoring the need for evidence-based antimicrobial strategies. This issue is further compounded by the finding that 45.2% of isolates originated from BSIs, where timely and effective therapy is critical to patient survival [42]. Alarmingly, resistance rates to commonly used empirical antibiotics were exceptionally high: 78% to third-generation cephalosporins and 71.6% to fluoroquinolones, indicating these agents can no longer be reliably recommended as first-line empirical treatments in our setting.

Our findings indicate that tigecycline have demonstrated 100% susceptibility across all tested isolates. However, its clinical application requires careful consideration due to several important factors. The limited penetration of tigecycline in BSIs poses a significant limitation [43]. This is especially important as the prevalence of *S. marcescens* was quite higher in blood cultures in this study. Furthermore, the need to preserve its effectiveness as one of the few remaining reliable options, coupled with the potential for development of resistance if used extensively, necessitates judicious use of this antimicrobial agent. For non-critical infections, aminoglycosides, particularly amikacin, which demonstrated the lowest resistance rate (49.6%) among its class, might be considered as part of combination therapy regimens. However, the effectiveness of empirical choices should be continuously evaluated through regular monitoring of local resistance patterns, allowing for timely adjustments to treatment protocols as needed.

While this study provides critical baseline data on CRSM in Pakistan, several limitations must be acknowledged. First, the absence of whole-genome sequencing (WGS) due to financial constraints restricted our ability to fully elucidate the genomic context of resistance genes. The WGS of these isolates will be helpful in clarifying the clonal relatedness of isolates and confirming the suspected outbreaks linked to $bla_{OXA-48}$ harboring strains and to map mobile genetic elements which are critical for understanding the dissemination of resistance across healthcare settings. Moreover, the geographic focus on the central Punjab region of Pakistan limits generalizability to other areas where infection control infrastructure may be weaker and antibiotic prescribing practice could be different. Furthermore, the lack of clinical outcome data (e.g., mortality, treatment failures) prevents direct correlation between resistance profiles and patient prognoses, which is essential for refining therapeutic guidelines. These limitations highlight the urgent need for investments in WGS infrastructure and expanded surveillance networks to address the escalating AMR crisis in the country.

## Conclusions

This first comprehensive study from Pakistan reveals a grim AMR landscape among *S. marcescens* marked by alarmingly high resistance to third-generation cephalosporins (78%) and carbapenems (52%). The widespread detection of critical resistance genes i.e. $bla_{OXA-48}$ (69.2%), *armA* (73.1%), and *aac*(6′)-Ib-cr (60.8%) among CRSM strains underscores the genetic complexity driving pan-resistance in *S. marcescens* isolates, leaving tigecycline as the sole susceptible agent. The predominance of bloodstream infections (45.2%), particularly in neonates and young patients, highlights CRSM as a lethal nosocomial pathogen in high-risk settings exacerbated by invasive medical practices and limited stewardship. The dissemination of $bla_{OXA-48}$ harboring CRSM strains necessitates urgent needs for stringent infection control to curb outbreaks. The study exposes the escalating AMR crisis in Pakistan fueled by unregulated antibiotic use and inadequate surveillance. The tailored stewardship is critical to mitigate this threat and prioritize rapid diagnostics to guide therapy. This baseline data calls for nationwide surveillance, genomic tracking of resistance mechanisms and policy reforms to enforce antibiotic guidelines. The findings serve as a warning and as well as a foundation for the AMR strategies in the region emphasizing that without urgent intervention CRSM-associated infections will surge thereby ruining global health security.

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## Declarations

**Conflict of interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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